Agentic AI-Driven Autonomous Covert Communication in MISO-NOMA VLC Systems With Imperfect CSI

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***Abstract*—This paper proposes an agentic artificial intelligence (AI) framework for autonomous covert communication in multiple-input single-output non-orthogonal multiple access (MISO-NOMA) visible light communication (VLC) systems. We conceptualize the transmitter as an autonomous Alice-agent that learns to establish covert links with a legitimate user (Bob) while remaining undetectable to an adversarial eavesdropper (Willie), despite lacking prior knowledge of the eavesdropper's exact channel state. To maintain stealth, the agent dynamically leverages artificial noise (AN) and public user signals as covers, making real-time decisions on power allocation and beamforming under imperfect channel state information. The problem is formulated as a partially observable Markov decision process (POMDP), and a novel transformer-based dual-clip proximal policy optimization (TDC-PPO) algorithm is introduced as the agent's decision-making engine. Specifically, the TDC-PPO agent utilizes a self-attention mechanism to enhance state representation and a dual-clipping strategy to ensure policy stability under adversarial conditions. Simulation results validate that the proposed agentic AI framework consistently achieves higher covert transmission rates than baseline deep reinforcement learning (DRL) methods while exhibiting superior adaptability and convergence across varying covertness constraints and quality-of-service requirements. This work validates the effectiveness of autonomous, learning-driven agents in governing complex, constraint-satisfying behaviors in secure wireless communications.**

***Index Terms*—Covert communication, VLC, agentic AI, NOMA, DRL.**

Received 22 January 2026; revised 25 March 2026; accepted 12 June 2026. Date of publication 22 June 2026; date of current version 1 July 2026. The work of Yangbo Guo and Ruihong Jiang were supported by the National Natural Science Foundation of China (NSFC) under Grant 62401201 and Grant 62301077. The work of Lei Zhang was supported by the Key R&D Projects in Henan Province under Grant 241111212800, and the First Batch of Leading Projects for AI + Education of Henan Provincial Department of Education under Grant 2024–10. The work of Baofang Chang was supported by the Science and Technology Research Project of Henan Province under Grant 252102520022. The associate editor coordinating the review of this article and approving it for publication was S. Zhang. *(Corresponding authors: Lei Zhang; Hao Deng.)*

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Digital Object Identifier 10.1109/TCCN.2026.3705848

I. Introduction

A. Background

THE rapid evolution of wireless technology, driven by the massive proliferation of the Internet of things (IoT) and data-intensive services, is continuously pushing the boundaries of beyond 5G (B5G) and 6G networks. Modern wireless networks must concurrently meet stringent requirements for ultra-high data rates, ubiquitous connectivity, robust security, and ultra-low latency. However, the available radio-frequency (RF) spectrum is increasingly congested, creating a critical bottleneck that limits the capacity to satisfy these escalating demands [1]. This spectrum crunch necessitates the exploration of innovative communication technologies. Visible light communication (VLC) has emerged as a highly promising candidate to supplement or potentially replace traditional RF links [2]. Leveraging the ubiquitous presence of light-emitting diode (LED) infrastructure, VLC uniquely offers the dual functionality of illumination and high-speed data transmission, making it exceptionally well-suited for indoor environments. Furthermore, VLC benefits from operating in a vast, unlicensed spectrum, inherently low inter-cell interference, and reduced infrastructure complexity [3]. In addition, VLC can be integrated with RF systems to form hybrid networks for improved energy efficiency and coverage [4], [5], and its security can be enhanced with simultaneous lightwave information and power transfer (SLIPT) techniques [6], [7].

While the inherent spatial confinement property of visible light prevents external eavesdroppers from intercepting information outside a room, it does not guarantee complete security. Paradoxically, the intrinsic broadcast nature of VLC, which is essential for uniform illumination and coverage, simultaneously creates a significant security vulnerability: malicious in-room eavesdroppers (Willie) can effortlessly capture data intended for legitimate receivers. This physical layer security (PLS) risk is particularly acute in large, shared indoor environments, such as smart offices or commercial spaces, where user communication security and privacy are paramount [8]. Consequently, investigating robust and innovative security paradigms for VLC systems is as critical as for their RF counterparts.

Against this backdrop, covert communication is emerging as a critical security paradigm that transcends conventional encryption methods [9]. The core objective of covert communication is to enable the reliable exchange of information between legitimate parties (Alice and Bob) while ensuring non-detection by an adversarial monitor (Willie). Unlike traditional PLS techniques that focus solely on protecting message confidentiality, covert communication provides communication deniability by concealing the very existence of the transmission. This robustness makes covert communication highly desirable for applications requiring the utmost secrecy and privacy. Regrettably, the vast majority of academic research on covert communication remains focused on traditional RF systems. The fundamental theory and practical realization for VLC are still largely unexplored [10]. Crucially, the substantial differences between VLC and RF—particularly concerning input signal constraints (non-negativity and linearity), channel models, and interference management—render existing RF-based covert communication solutions directly inapplicable to the optical domain. Consequently, new foundational principles must be established for VLC-based covert communication, especially when considering the complexity of modern multi-user architectures.

In parallel, non-orthogonal multiple access (NOMA) has emerged as an indispensable multiple-access scheme for next-generation wireless systems, capable of simultaneously addressing the demands for massive connectivity and high spectral efficiency [11]. This is achieved by relying on superposition coding at the transmitter, where signals for different users (e.g., Bob and the public user Carol) are superimposed and transmitted using distinct power allocation coefficients [12]. At the receiver side, successive interference cancellation (SIC) is employed to sequentially decode and subtract the stronger signals, recovering the desired low-power signal. The performance of NOMA critically hinges upon precise power allocation. When integrated with multiple LED transmitters in a multiple-input single-output (MISO) configuration, this forms a MISO-NOMA VLC system. Accordingly, this paper investigates resource optimization for covert transmission within the MISO-NOMA VLC architecture. The complex interplay between power allocation and the inherent security constraints underscores the necessity of a sophisticated strategy to ensure both the reliability of legitimate transmission and the robustness of covertness.

B. Related Work

The integration of PLS into VLC has been widely recognized as a critical research direction. Authorities like the Visible Light Communication Consortium (VLCC) have acknowledged the necessity of robust security solutions due to the intrinsic broadcast nature of VLC, which makes information susceptible to in-room eavesdroppers [13], [14], [15]. The majority of existing PLS research in VLC has traditionally focused on two main streams. The first stream concentrates on maximizing the secrecy capacity, which serves as the key performance metric for analyzing system security. Early studies in this area focused on deriving secrecy rate bounds for single-input single-output (SISO) wiretap VLC channels [16]. More practically relevant works have since progressed to MISO scenarios, leveraging techniques like beamforming and resource allocation to enhance the legitimate channel while suppressing the eavesdropper's channel [17]. However, these PLS works inherently fail to address a more fundamental security demand: communication deniability. Since their primary goal is protecting the message content, they remain detectable by a monitoring party, which is insufficient for applications requiring the concealment of the transmission act itself.

The need for high spectral efficiency has driven extensive research into NOMA, a key enabler for massive connectivity [18]. Consequently, NOMA-based covert communication has become a prominent academic focus, largely because the superimposed transmission inherent in NOMA can be leveraged to shield covert activities [19]. Researchers have explored various approaches to maximize covert throughput in NOMA systems. For instance, studies have focused on using non-covert transmissions to dynamically cover the covert link [20], [21], optimizing the trajectory of mobile relays to enhance covert capacity [22], and utilizing energy-harvesting jammers to disrupt the monitor's detection process [23], [24], While these studies validate the potential of NOMA to enhance covert communication performance, they overwhelmingly focus on the RF domain. The distinct characteristics of VLC, particularly its channel behavior and signal amplitude constraints, mean that these RF-centric NOMA-based covert communication solutions are not directly transferable to the optical domain.

Addressing the limitations of RF studies, a smaller body of work has explored covert communication specifically in the VLC domain. The work in [25] investigated a UAV-based VLC system and derived the optimal flight altitude for covertness. Furthermore, advanced reconfigurable intelligent surfaces (RIS) have been proposed to optimize the channel for covert light transmission [26]. Parallel to these efforts, robust covert communication design has emerged to tackle the critical challenge of non-ideal channel state information (CSI), which is endemic to practical networks [27]. Various techniques have been proposed to maximize the covert transmission rate under CSI uncertainty, including robust beamforming [28] and intelligent control of RIS to mitigate information leakage [29], [30]. Recently, generative AI has been leveraged to address CSI uncertainty in physical layer authentication, demonstrating the potential of adaptive fingerprint generation for robust security in dynamic environments [31]. However, our review reveals a significant research gap: no existing work has simultaneously investigated the performance of NOMA-based covert communication in VLC systems under the constraint of non-ideal CSI. This intersection represents a complex, multi-constrained optimization problem that requires a paradigm shift beyond conventional methods [32].

The inherent complexities of joint power allocation and beamforming in MISO-NOMA VLC systems—compounded by the integration of artificial noise (AN) to mask transmissions, stringent covertness requirements, and physical hardware constraints under CSI uncertainty—result in a highly non-convex and intractable optimization landscape. Traditional secure communication technologies primarily focus on preventing information content from being eavesdropped upon [33]; however, the rapid evolution of monitoring technologies renders even encrypted data vulnerable to sophisticated detection and traffic analysis [34]. Covert communication shifts the paradigm from content protection to existence concealment [35], necessitating the Alice-agent to proactively manipulate the environment by injecting dynamic AN to obfuscate signal presence.

Conventional optimization techniques rely on iterative approximations that are hypersensitive to initial conditions and the stochastic nature of VLC channels, frequently failing to guarantee real-time convergence or global optimality. Standard deep reinforcement learning frameworks, while capable of handling sequential decision making, are not inherently designed for such tightly coupled multi objective constraints and often struggle with the harsh negative rewards induced by covertness violations, leading to unstable policy updates and inefficient exploration in safety critical regimes. Recent work has integrated mixture of experts architectures with proximal policy optimization (PPO) to alleviate learning complexity and mitigate gradient interference in multi objective wireless network optimization [36], [37]. Similarly, the work in [38] demonstrates the effectiveness of PPO in maximizing energy efficiency for RSMA-IRS-assisted integrated sensing and communication systems under realistic channel models. The adversarial nature of this environment fundamentally demands a sequential decision making framework capable of autonomous learning and instantaneous adaptation, a level of operational autonomy and multidimensional complexity that conventional optimization or static machine learning models cannot fully address [39]. This leads us to adopt an agentic AI approach, where the Alice agent is not merely a policy optimizer but an autonomous entity that perceives its surroundings, interprets partial observations, makes sequential decisions under uncertainty, and continuously refines its strategies to balance multiple conflicting objectives such as covertness, quality of service, and hardware constraints. This self directed behavior, particularly in the face of unknown eavesdropper channels and real time feedback, distinguishes the proposed agentic framework from standard DRL implementations, embodying the core principles of agentic artificial intelligence.

C. Motivation and Contributions

Despite the substantial progress in physical-layer security and covert communication, a critical gap persists in addressing the symbiotic challenges of MISO-NOMA VLC systems under non-ideal channel conditions. In such architectures, maximizing the covert rate necessitates a sophisticated joint optimization of power allocation and beamforming. This process is uniquely complicated by the integration of AN, which acts as a dynamic obfuscation layer to shield covert transmissions. Managing these intricately coupled constraints—ranging from communication deniability and public user QoS to the strict linear operating range of LEDs—results in an optimization landscape that is both highly non-convex and mathematically intractable.

Furthermore, the inherent uncertainty of the eavesdropper's CSI creates a volatile adversarial environment where conventional, static optimization techniques fail to guarantee robust performance or real-time convergence. This necessitates a shift toward an intelligent, adaptive resource allocation paradigm capable of autonomous decision-making. While deep reinforcement learning (DRL) has demonstrated efficacy in solving sequential decision-making problems, the multi-objective nature of our framework—balancing effective masking via AN with reliable NOMA-based data delivery—demands an Agentic AI approach. By adopting an agentic DRL-based framework, this study enables the Alice-agent to proactively learn and navigate the trade-offs between information throughput and non-detectability, thereby providing a resilient and high-performance solution for next-generation secure VLC networks.

The main contributions of this paper are summarized as follows:

- We propose a novel agentic AI framework where the MISO transmitter operates as an autonomous Alice-agent, capable of orchestrating its signal and masking resources. Within this framework, we formulate a joint power allocation and AN beamforming optimization problem to maximize the covert rate. This problem is designed to navigate a complex multi-constraint landscape, encompassing the covertness requirement (bolstered by active AN-aided obfuscation), the public user's QoS threshold, the total power budget, NOMA SIC prerequisites, and the linear operating range of the LEDs.
- Recognizing the challenge as a mixed-integer nonlinear programming (MINLP) problem, we first reformulate the optimization as a partially observable Markov decision process (POMDP). We then introduce a novel transformer-based dual-clipped proximal policy optimization (TDC-PPO) algorithm as the agent's decision engine. This approach leverages a self-attention mechanism for enhanced environmental state representation and a dual-clipping strategy to ensure stable and robust policy updates in adversarial and constrained environments. We utilize generalized advantage estimation (GAE) to refine policy gradient stability.
- Through comprehensive simulations, we demonstrate that the proposed TDC-PPO algorithm achieves excellent convergence performance and significantly outperforms existing baseline methods in terms of achievable covert rate. We further analyze the impact of various practical parameters—such as LED height, QoS thresholds, covertness constraints, power budgets, and monitor proximity—on the overall system performance, providing valuable design insights for MISO-NOMA VLC networks.

The remainder of this paper is organized as follows. Section II establishes the NOMA-MISO VLC system model and formulates the covert communication optimization problem. In Section III, we detail the proposed Agentic AI framework and reformulate the resource allocation task as

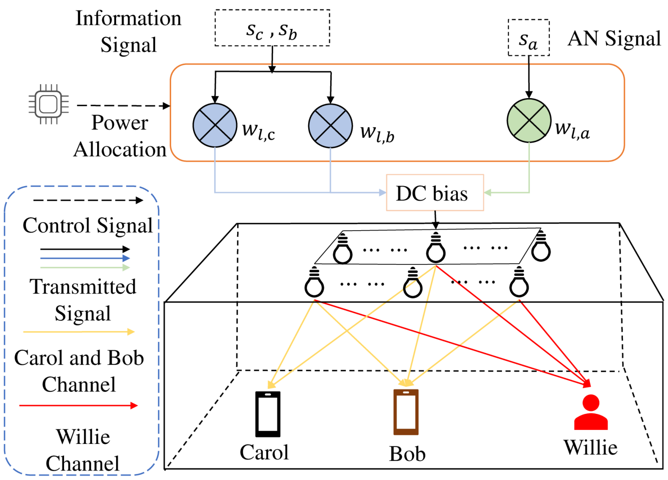

Fig. 1. Architectural framework of the Agentic AI-enabled covert VLC system, where the Alice-agent autonomously optimizes transmission strategies to balance the QoS of public user Carol and the covertness of Bob against the adversarial observer Willie.

a POMDP. Section IV presents the technical design of the proposed TDC-PPO algorithm, including its self-attention mechanism and dual-clipping strategy. Simulation results and comprehensive performance analyses are provided in Section V, and the paper is concluded in Section VI.

II. System Model and Problem Formulation

A. Channel and Signal Models

The investigated covert communication system operates within a MISO-NOMA VLC system framework. As depicted in Fig. 1, the system comprises a legitimate transmitter (Alice-agent) equipped with L LEDs; a covert user (Bob); a public user (Carol); and an adversarial eavesdropper (Willie). Unlike conventional RF systems where beamforming uses phase adjustments, VLC relies on intensity modulation and direct detection, so beamforming is achieved via power allocation across LEDs and the beamforming vectors are real and nonnegative. The Alice-agent is tasked with transmitting a composite signal, integrating the information of Bob and Carol, and a generated AN. In this framework, the covertness is doubly reinforced: Carol's public transmission provides a coarse cover, while the AN provides fine-grained stochastic obfuscation. This AN is not merely a jamming signal; it is an active masking tool designed to exploit the channel uncertainty at Willie, effectively drowning Bob's signal power in a controlled noise floor. For simplicity, all receivers use a single vertically oriented photodiode (PD). A DC-bias $\mathbf{I}_{\mathrm{DC}}$ is applied to ensure constant LED illumination, preventing detection via visual observation.

Each VLC link comprises both line-of-sight (LOS) and non-line-of-sight (NLOS) components. However, only the LOS component is considered in this analysis, as it typically accounts for 95% of the total received power at the PDs. Under the Lambertian radiation model for LED illumination, the direct LOS channel gain from the l-th LED to Carol is given by

$$h_{l,\mathrm{c}} = \frac{(m+1)\,A_r}{2\pi D_{l,\mathrm{c}}^2} T_f g_{\mathrm{of}} \cos^m\left(\Phi_{l,\mathrm{c}}\right)\cos\left(\Psi_{l,\mathrm{c}}\right), \tag{1}$$

where A_r denotes the PD physical area, m is the Lambertian order, and $D_{l,\mathrm{c}}$ is the distance between the l-th LED and Carol. $\Phi_{l,\mathrm{c}}$ and $\Psi_{l,\mathrm{c}}$ represent the emission and incidence angle, respectively. T_f is the optical concentrator gain, which depends on the field-of-view (FoV) half-angle, and g_{of} is the optical filter gain. The direct LoS channel gains $h_{l,\mathrm{b}}$ (to Bob) and $h_{l,\mathrm{w}}$ (to Willie) are also modeled using the same Lambertian formulation.

The CSI for the legitimate Alice-Carol and Alice-Bob links is assumed to be perfectly known a priori due to the cooperative relationship. In contrast, the CSI of the Alice-Willie channel, $h_{l,\mathrm{w}}$, is subject to uncertainty. This uncertainty is modeled as $h_{l,\mathrm{w}} = \widehat{h}_{l,\mathrm{w}} + \Delta h_{l,\mathrm{w}}$, where $\widehat{h}_{l,w} \sim \mathcal{N}(0,(1-\kappa)\sigma^2)$ is Willie's estimated channel and $\Delta h_{l,w} \sim \mathcal{N}(0,\kappa\sigma^2)$ is the estimation error, with $\kappa \in [0,1]$ being the uncertainty factor. The LoS channel gains are expressed as vectors: $\mathbf{h}_{\mathrm{c}} = [h_{1,\mathrm{c}},\ldots,h_{L,\mathrm{c}}]^T \in \mathbb{R}_+^{L\times 1}$ for Carol, $\mathbf{h}_{\mathrm{b}} = [h_{1,\mathrm{b}},\ldots,h_{L,\mathrm{b}}]^T \in \mathbb{R}_+^{L\times 1}$ for Bob, and $\mathbf{h}_{\mathrm{w}} = [h_{1,\mathrm{w}},\ldots,h_{L,\mathrm{w}}]^T \in \mathbb{R}_+^{L\times 1}$ for Willie.

Let s_c and s_b represent the optical pulse amplitude modulation (PAM) symbols for Carol and Bob, respectively, and s_a denote the AN signal. Without loss of generality, s_c, s_b, and s_a are zero-mean and uniformly distributed over $[-1,1]$. To ensure the transmitted optical signal is non-negative, a DC-bias vector $\mathbf{I}_{\mathrm{DC}}$ is applied. The resulting composite signal vector at the i-th symbol period is formulated as

$$\begin{aligned}\mathbf{x}[i] &= [x_1[i],\ldots,x_L[i]]^T\\ &= \mathbf{w}_{\mathrm{c}}s_c[i] + \mathbf{w}_{\mathrm{b}}s_b[i] + \mathbf{w}_{\mathrm{a}}s_a[i] + \mathbf{I}_{\mathrm{DC}}[i],\end{aligned} \tag{2}$$

where $\mathbf{w}_{\mathrm{c}}, \mathbf{w}_{\mathrm{b}}, \mathbf{w}_{\mathrm{a}} \in \mathbb{R}_+^{L\times 1}$ are the beamforming vectors for Carol's information, Bob's covert information, and the active masking AN, respectively. Therefore, the total electrical power budget is constrained by $\|\mathbf{w}_{\mathrm{c}}\|^2 + \|\mathbf{w}_{\mathrm{b}}\|^2 + \|\mathbf{w}_{\mathrm{a}}\|^2 \le P_{\max}$.

Each LED exhibits a linear dynamic range where the optical output scales proportionally with the driving current x_l. To suppress nonlinear distortion and ensure stable illumination, the signal should satisfy

$$I_{\min} \le x_l \le I_{\max}, \quad \forall l \in \mathcal{L}. \tag{3}$$

Given the bounded nature of these signal symbols, the peak power allocation per LED is strictly governed by

$$w_{l,\mathrm{c}} + w_{l,\mathrm{b}} + w_{l,\mathrm{a}} \le \delta_{\mathrm{DC}}, \tag{4}$$

where $\delta_{\mathrm{DC}} = \min\{I_{\mathrm{DC}} - I_{\min}, I_{\max} - I_{\mathrm{DC}}\}$ represents the maximum allowable AC signal swing. Consequently, the signals received at Carol, Bob, and the adversarial monitor Willie are expressed as

$$y_{\mathrm{c}}[i] = \mathbf{h}_{\mathrm{c}}^T(\mathbf{w}_{\mathrm{c}}s_c[i] + \mathbf{w}_{\mathrm{b}}s_b[i] + \mathbf{w}_{\mathrm{a}}s_a[i]) + n_{\mathrm{c}}[i], \tag{5}$$

$$y_{\mathrm{b}}[i] = \mathbf{h}_{\mathrm{b}}^T(\mathbf{w}_{\mathrm{c}}s_c[i] + \mathbf{w}_{\mathrm{b}}s_b[i] + \mathbf{w}_{\mathrm{a}}s_a[i]) + n_{\mathrm{b}}[i], \tag{6}$$

$$y_{\mathrm{w}}[i] = \mathbf{h}_{\mathrm{w}}^T(\mathbf{w}_{\mathrm{c}}s_c[i] + \mathbf{w}_{\mathrm{b}}s_b[i] + \mathbf{w}_{\mathrm{a}}s_a[i]) + n_{\mathrm{w}}[i], \tag{7}$$

where $n_k[i] \sim \mathcal{N}(0,\sigma^2)$ $(k \in \{\mathrm{c},\mathrm{b},\mathrm{w}\})$ denotes the additive white Gaussian noise (AWGN) at each respective receiver.

B. NOMA Decoding and Covertness Analysis

Assuming $\|\mathbf{h}_{\mathrm{b}}\|^2 \geq \|\mathbf{h}_{\mathrm{c}}\|^2$, legitimate users employ successive interference cancellation (SIC) to eliminate the AN and interference. The AN is assigned the highest power priority to enable decoding first, followed by s_c and s_b. The resulting achievable rates at Bob for the detection of s_c and s_b are formulated as

$$R_{\mathrm{b}}^c = \frac{1}{2}\log_2\left(1+\frac{e}{2\pi}\gamma_{\mathrm{b}}^c\right), \quad \gamma_{\mathrm{b}}^c = \frac{(\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{c}})^2}{(\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{b}})^2+\sigma^2}, \tag{8}$$

$$R_{\mathrm{b}}^b = \frac{1}{2}\log_2\left(1+\frac{e}{2\pi}\gamma_{\mathrm{b}}^b\right), \quad \gamma_{\mathrm{b}}^b = \frac{(\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{b}})^2}{\sigma^2}. \tag{9}$$

Correspondingly, the achievable rate for Carol is given by

$$R_{\mathrm{c}}^c = \frac{1}{2}\log_2\left(1+\frac{e}{2\pi}\gamma_{\mathrm{c}}^c\right), \quad \gamma_{\mathrm{c}}^c = \frac{(\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{c}})^2}{(\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{b}})^2+\sigma^2}. \tag{10}$$

The adversarial monitor, Willie, performs binary hypothesis testing based on his received signal power to detect the presence of the covert transmission s_b. The two competing hypotheses are defined as

$$\mathcal{H}_0: y_{\mathrm{w}}[i] = \mathbf{h}_{\mathrm{w}}^T(\mathbf{w}_{\mathrm{c}}s_c[i]+\mathbf{w}_{\mathrm{a}}s_a[i]) + n_{\mathrm{w}}[i], \tag{11}$$

$$\mathcal{H}_1: y_{\mathrm{w}}[i] = \mathbf{h}_{\mathrm{w}}^T(\mathbf{w}_{\mathrm{c}}s_c[i]+\mathbf{w}_{\mathrm{b}}s_b[i]+\mathbf{w}_{\mathrm{a}}s_a[i]) + n_{\mathrm{w}}[i]. \tag{12}$$

Under $\mathcal{H}_0$, Willie observes only the public NOMA transmission and the masking AN, whereas under $\mathcal{H}_1$, the covert signal s_b is additionally embedded within the composite transmission. To quantify the detectability of the covert link, we assume that Willie employs the Neyman-Pearson criterion for binary hypothesis testing. The optimal decision rule that minimizes the Detection Error Probability (DEP) is the likelihood ratio test, formulated as

$$\frac{p_1(y_{\mathrm{w}})}{p_0(y_{\mathrm{w}})} \underset{D_0}{\overset{D_1}{\gtrless}} 1, \tag{13}$$

where D_0 and D_1 denote Willie's binary decisions favoring hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively. Under the Gaussian signal assumption, the likelihood functions are given by $p_0(y_{\mathrm{w}}) = \frac{1}{\pi\lambda_0}\exp\left(-\frac{|y_{\mathrm{w}}|^2}{\lambda_0}\right)$ and $p_1(y_{\mathrm{w}}) = \frac{1}{\pi\lambda_1}\exp\left(-\frac{|y_{\mathrm{w}}|^2}{\lambda_1}\right)$. Here, the average received power terms are defined as $\lambda_0 = \left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{c}}\right|^2 + \left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{a}}\right|^2 + \sigma^2$ and $\lambda_1 = \left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{b}}\right|^2+\left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{c}}\right|^2+\left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{a}}\right|^2+\sigma^2$. In practice, the likelihood ratio test is equivalent to comparing the average received signal power over N symbols against an optimal threshold τ:

$$P_{\mathrm{w}} = \frac{1}{N}\sum_{i=1}^{N}|y_{\mathrm{w}}[i]|^2 \underset{\mathcal{D}_0}{\overset{\mathcal{D}_1}{\gtrless}} \tau, \tag{14}$$

where $\tau = \frac{\lambda_0\lambda_1}{\lambda_1-\lambda_0}\ln\frac{\lambda_1}{\lambda_0}$ is the optimal detection threshold satisfying $\tau > 0$. The resulting minimum DEP achievable by Willie is

$$\begin{aligned}\xi_{\mathrm{w}} &= P_{\mathrm{FA}} + P_{\mathrm{MD}} \\ &= \Pr\left(|y_{\mathrm{w}}|^2 \geq \tau \mid \mathcal{H}_0\right) + \Pr\left(|y_{\mathrm{w}}|^2 \leq \tau \mid \mathcal{H}_1\right) \\ &= 1 + \left(\frac{\lambda_1}{\lambda_0}\right)^{-\frac{\lambda_1}{\lambda_1-\lambda_0}} - \left(\frac{\lambda_1}{\lambda_0}\right)^{-\frac{\lambda_0}{\lambda_1-\lambda_0}},\end{aligned} \tag{15}$$

where $P_{\mathrm{FA}} = \Pr(D_1 \mid \mathcal{H}_0)$ and $P_{\mathrm{MD}} = \Pr(D_0 \mid \mathcal{H}_1)$ denote the false alarm and missed detection probabilities, respectively.

However, the non-convex closed-form expression of ξ_{w} is mathematically intractable for direct optimization. To circumvent this, we employ Pinsker's inequality to establish a more tractable lower bound:

$$\xi_{\mathrm{w}} \geq 1 - \sqrt{\frac{1}{2}D(p_0\|p_1)}, \tag{16}$$

where $D(p_0\|p_1) = \int p_0(y_{\mathrm{w}})\ln\frac{p_0(y_{\mathrm{w}})}{p_1(y_{\mathrm{w}})}dy_{\mathrm{w}} = \ln\frac{\lambda_1}{\lambda_0} + \frac{\lambda_0}{\lambda_1} - 1$ is the Kullback-Leibler (KL) divergence between the distributions. To guarantee a predefined level of covertness, Willie's minimum DEP must satisfy $\xi_{\mathrm{w}} \geq 1-\epsilon$, where $\epsilon > 0$ represents the required covertness level. Combining this requirement with the lower bound yields a more conservative and tractable covertness constraint, i.e., $D(p_0\|p_1) \leq 2\epsilon^2$. Let $f(x) = \ln x + \frac{1}{x} - 1$ for $x \in [1, +\infty)$. Given that $f(x) = 2\epsilon^2$ has a unique solution $\omega \in [1, +\infty)$, and leveraging the monotonicity of $f(x)$, the covertness constraint can be equivalently expressed as

$$\left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{b}}\right|^2 \leq (\omega-1)\left(\left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{c}}\right|^2 + \left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{a}}\right|^2 + \sigma^2\right). \tag{17}$$

C. Optimization Problem Formulation

The design goal of this study is to maximize the covert transmission rate R_{b}^b through the joint optimization of power allocation and transmit beamforming vectors $\{\mathbf{w}_{\mathrm{c}}, \mathbf{w}_{\mathrm{b}}, \mathbf{w}_{\mathrm{a}}\}$. This maximization is performed under the governance of the Alice-agent to ensure stringent adherence to covertness requirements, Carol's QoS thresholds, and the physical hardware limitations of the VLC system. Integrating the previously derived models, the optimization problem is formulated as

$$(\mathbf{P1}): \quad \max_{\mathbf{w}_{\mathrm{c}},\mathbf{w}_{\mathrm{b}},\mathbf{w}_{\mathrm{a}}} R_{\mathrm{b}}^b \tag{18a}$$

$$\text{s.t.} \quad \xi_{\mathrm{w}} \geq 1-\epsilon, \tag{18b}$$

$$R_{\mathrm{c}}^c \geq R_{\mathrm{c}}^{th}, \tag{18c}$$

$$\|\mathbf{w}_{\mathrm{c}}\|^2 + \|\mathbf{w}_{\mathrm{b}}\|^2 + \|\mathbf{w}_{\mathrm{a}}\|^2 \leq P_{\max}, \tag{18d}$$

$$w_{l,\mathrm{c}} + w_{l,\mathrm{b}} + w_{l,\mathrm{a}} \leq \delta_{\mathrm{DC}}, \tag{18e}$$

$$\left|\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{c}}\right|^2 \geq \left|\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{b}}\right|^2, \ \left|\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{c}}\right|^2 \geq \left|\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{b}}\right|^2, \tag{18f}$$

$$\left|\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{a}}\right|^2 \geq \left|\mathbf{h}_{\mathrm{c}}^T\mathbf{w}_{\mathrm{c}}\right|^2, \ \left|\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{a}}\right|^2 \geq \left|\mathbf{h}_{\mathrm{b}}^T\mathbf{w}_{\mathrm{c}}\right|^2, \tag{18g}$$

$$\|\mathbf{w}_{\mathrm{c}}\|^2 \geq \|\mathbf{w}_{\mathrm{b}}\|^2, \ R_{\mathrm{b}}^c \geq R_{\mathrm{c}}^c. \tag{18h}$$

In **P1**, constraint (18b) defines the covertness requirement, ensuring that the detection error at Willie remains above the predefined threshold. Constraint (18c) maintains the QoS of the Carol by enforcing a minimum target rate R_{c}^{th}. The electrical power limitations are addressed in (18d) for the total transmit power and in (18e) for the per-LED linear dynamic range. Furthermore, (18f) ensures rate fairness and effective signal distinction between users, while (18g) is specifically designed to manage the AN. It ensures that the AN power is sufficient for masking purposes yet structured to be successfully canceled by legitimate users via SIC. Finally, (18h) guarantees the decodability of the NOMA protocol, enforcing the necessary power and rate relationships for successful successive interference cancellation.

III. Agentic AI Framework and DRL Formulation

This section reformulates the static optimization problem **P1** into a dynamic, autonomous decision-making paradigm. We first outline the fundamental principles of DRL and then formally map the physical constraints of the MISO-NOMA VLC system into the state, action, and reward spaces of an Alice-agent.

A. The Alice-Agent and POMDP Environment

We conceptualize the transmitter as an autonomous Alice-agent operating under strictly bounded rationality. Unlike conventional schemes that assume perfect global CSI, the agent must make decisions based on local observations and imperfect estimates of the adversarial channel $\mathbf{h}_{\mathrm{w}}$.

The environment is characterized by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma \rangle$. At each time step t, the agent observes a state s_t and executes an action a_t according to a stochastic policy $\Omega_\theta(a_t|s_t)$, Here, Ω_θ represents a conditional probability distribution over the action space $\mathcal{A}$, parameterized by a deep neural network with weights θ. The system then transitions to s_{t+1} based on the state transition probability $\mathcal{P}(s_{t+1}|s_t, a_t)$, yielding an immediate reward r_t. The term $\gamma \in (0,1)$ is the discount factor, which determines the agent's horizon by balancing immediate rewards against future gains.

B. State Space Formulation

The state space $\mathcal{S}$ must capture the temporal dependencies of the channel and the immediate performance feedback. We define the state vector $s_t \in \mathcal{S}$ as

$$s_t = \left(\mathbf{a}_{t-1}, r_{t-1}, R_{\mathrm{c}}^c, R_{\mathrm{b}}^c, R_{\mathrm{b}}^b\right), \tag{19}$$

where $\mathbf{a}_{t-1}$ is the action from the previous time step, and r_{t-1} is the received reward. The terms R_{c}^c, R_{b}^c, and R_{b}^b represent the instantaneous achievable rates for the public and covert users. This $(3L+4)$-dimensional vector provides the agent with essential context regarding the "safety" of its recent power allocation strategy relative to the dynamic channel conditions.

C. Action Space Formulation

The action space $\mathcal{A}$ corresponds to the continuous power allocation variables. At each step, the Alice-agent outputs the beamforming vectors for the public signal, the covert signal, and the strategic AN

$$a_t = [\mathbf{w}_{\mathrm{c}}, \mathbf{w}_{\mathrm{b}}, \mathbf{w}_{\mathrm{a}}] \in \mathcal{A}, \tag{20}$$

where each $\mathbf{w}_k \in \mathbb{R}^{L\times 1}$. This results in a continuous action space of dimension $3L$. The inclusion of $\mathbf{w}_{\mathrm{a}}$ as an active decision variable allows the agent to dynamically modulate the jamming intensity to mask covert transmissions.

D. Reward Function as a Safety Logic Gate

To impart goal-directed behavior, we design the reward function as a strict safety logic gate. This ensures that the agent receives a positive reward proportional to the covert rate R_{b}^b if and only if all system constraints are simultaneously satisfied. The immediate reward r_t is defined as:

$$r_t = R_{\mathrm{b}}^b \times \Gamma_{\text{Cov}} \times \Gamma_{\text{QoS}} \times \Gamma_{\text{Pwr}} \times \Gamma_{\text{Lin}} \times \Gamma_{\text{Fair}} \times \Gamma_{\text{AN}} \times \Gamma_{\text{SIC}}, \tag{21}$$

where the binary indicators $\Gamma_i \in \{0,1\}$ penalize violations of the constraints derived in problem **P1**:

- Covertness(Γ_{Cov}): Enforces $\frac{\left|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{b}}\right|^2}{\left(|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{c}}|^2+|\mathbf{h}_{\mathrm{w}}^T\mathbf{w}_{\mathrm{a}}|^2+\sigma^2\right)} \leq \omega - 1$, preventing detection by Willie.
- QoS (Γ_{QoS}): Ensures $R_{\mathrm{c}}^c \geq R_{\mathrm{c}}^{\mathrm{th}}$ for the public user.
- Hardware limits ($\Gamma_{\text{Pwr}}, \Gamma_{\text{Lin}}$): Enforce the total power budget $P_{\max}$ and per-LED operation linearity δ_{DC}.
- NOMA protocol ($\Gamma_{\text{Fair}}, \Gamma_{\text{SIC}}$): Guarantee the power ordering required for successful SIC decoding.
- AN efficacy (Γ_{AN}): Ensures AN is sufficient to mask signals yet cancelable by legitimate users, as defined in constraint (18g).

This multiplicative structure forces the agent to learn a policy that inherently prioritizes constraint satisfaction over greedy rate maximization.

E. Policy Optimization Objective

The agent seeks the optimal policy parameters θ^* that maximize the expected discounted cumulative reward $J(\Omega_\theta)$:

$$J(\Omega_\theta) = \mathbb{E}_{\tau\sim\Omega_\theta}\left[\sum_{t=0}^{\infty}\gamma^t r_t(s_t, a_t)\right], \tag{22}$$

where $\tau = (s_0, a_0, r_0, s_1, \ldots)$ denotes a trajectory sampled under policy Ω_θ, and the expectation is taken over the stationary state-action distribution ρ_{Ω_θ}. The optimization problem is formulated as

$$\begin{aligned} \max_{\Omega_\theta} \quad & J(\Omega_\theta) \\ \text{s.t.} \quad & a_t \sim \Omega_\theta(a_t|s_t), \\ & s_{t+1} \sim \mathcal{P}(s_{t+1}|s_t, a_t). \end{aligned} \tag{23}$$

Maximizing $J(\Omega_\theta)$ is challenging due to the coupling of non-convex constraints and the absence of Willie's instantaneous CSI. To address this, we propose the TDC-PPO algorithm, detailed in the following section.

IV. The Proposed TDC-PPO Algorithm

To enable the Alice-agent to operate autonomously within the adversarial and partially observable VLC environment defined in Section III, we develop a specialized learning architecture. This framework serves as the agent's "cognitive engine", integrating high-dimensional perception, stochastic decision-making, and robust adaptation. Specifically, we propose the TDC-PPO algorithm, the architecture of which is illustrated in Fig. 2.

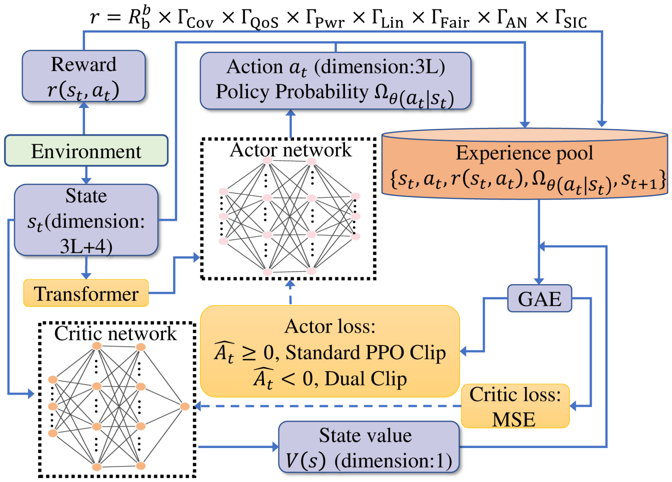

Fig. 2. Architecture of the TDC-PPO agent. The agent perceives the environment state, uses a transformer module for state representation, and outputs an action through its actor network, which is trained with a dual-clip mechanism for stability.

A. Situational Awareness via Self-Attention Encoding

A fundamental challenge for an autonomous agent is to distill actionable insights from high-dimensional, sequential, or structured environmental data. Traditional feed-forward networks often struggle to capture the long-range interdependencies within the state vector $s_t = (\mathbf{a}_{t-1}, r_{t-1}, R_{\mathrm{c}}^{c}, R_{\mathrm{b}}^{c}, R_{\mathrm{b}}^{b})$, where temporal correlations and multi-user rate couplings are critical.

To achieve superior situational awareness, our agent employs a self-attention mechanism to encode its observations. By dynamically weighing different components of the state history, the agent can prioritize features most relevant to its current covertness risk. Specifically, for a given hidden state vector X, the Query (Q), Key (K), and Value (V) representations are respectively computed as

$$Q = XW_Q, \tag{24}$$

$$K = XW_K, \tag{25}$$

$$V = XW_V, \tag{26}$$

where W_Q, W_K, and W_V are learnable weight matrices. Subsequently, The context-aware state representation Z_t is then derived via the scaled dot-product attention:

$$Z_t = \mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{27}$$

where d_k denotes the dimension of the key vectors. This attention-driven perception mimics an internal reasoning process, allowing the agent to infer, for instance, that a marginal decline in R_{b}^{b} following a specific power adjustment likely signals increased detection scrutiny by Willie, thereby necessitating a more conservative policy in subsequent steps.

B. Stochastic Decision-Making and Policy Mapping

The processed state representation Z_t is fed into a multilayer perceptron (MLP), which outputs the parameters of a probability distribution over the action space — the agent's policy $\Omega_\theta(a_t|Z_t)$. In the continuous action space of VLC power allocation, defined by $a_t = [\mathbf{w}_{\mathrm{c}}, \mathbf{w}_{\mathrm{b}}, \mathbf{w}_{\mathrm{a}}]$, the network outputs the mean μ_t and covariance $\sigma_t^2\mathbf{I}$ of a multivariate Gaussian distribution. The agent then samples its action as $a_t \sim \mathcal{N}(\mu_t, \sigma_t^2\mathbf{I})$. This stochasticity is indispensable for exploration, enabling the agent to probe the boundaries of the covert-rate region and discover optimal beamforming strategies under fluctuating CSI and adversarial conditions.

C. Robust Learning via Dual-Clip PPO

Learning from experience is what transforms a static policy into an adaptive, intelligent agent. We adopt a policy gradient approach where the agent updates its policy parameters θ to maximize the expected cumulative reward $J(\Omega_\theta)$. The standard PPO algorithm employs a clipped surrogate objective function to restrict the magnitude of policy updates. In each training iteration, the algorithm collects a batch of trajectories by interacting with the environment using the current policy. Subsequently, the advantage function $\hat{A}_t$, which quantifies the desirability of an action in a given state, is estimated via generalized advantage estimation (GAE). The objective function for the policy update in standard PPO is

$$L^{\mathrm{CLIP}}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{28}$$

where $r_t(\theta) = \frac{\Omega_\theta(a_t|s_t)}{\Omega_{\theta_{\mathrm{old}}}(a_t|s_t)}$ is the probability ratio between the new and old policies. clip$(\cdot)$ operation constrains the update step size, thereby preventing instability caused by excessively large policy changes.

However, in adversarial VLC environments, the agent frequently encounters negative advantages ($\hat{A}_t < 0$), where an action results in a severe covertness violation or zero reward. Standard PPO can suffer from training divergence when a low-probability action yields a large negative advantage, leading to catastrophic policy shifts. To ensure dependable performance, we introduce the dual-clip PPO mechanism that establishes a safety floor for policy degradation [40]. The comprehensive policy loss function, which incorporates this dual-clipping logic to stabilize learning under negative advantages, is formulated in (29), as shown at the bottom of the page.

As articulated in (29), while the agent continues to reinforce beneficial actions ($\hat{A}_t \geq 0$), the additional lower-bound clip factor $c > 1$ constrains the penalty magnitude when $\hat{A}_t < 0$. The hyperparameter c directly controls how much a large probability ratio can worsen the policy when an action yields a strongly negative advantage. Setting c too close to 1 offers little protection against catastrophic updates, whereas

$$L(\theta) = \mathbb{E}_t\left[\begin{cases}\min\left(r_t(\theta)\hat{A}_t, \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right), & \text{if } \hat{A}_t \geq 0\\ \max\left(\min\left(r_t(\theta)\hat{A}_t, \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right), c\hat{A}_t\right), & \text{if } \hat{A}_t < 0\end{cases}\right] \tag{29}$$

Algorithm 1 TDC-PPO Agent Training Loop

Require: Initial policy Ω_θ with self-attention, value network V_ϕ, clip factor ϵ, dual-clip factor c, discount factor γ, GAE parameter λ, batch size B, epochs K, total iterations M.
Ensure: Optimized policy network Ω_θ

```
Initialize parameters θ, φ, and old policy θ_old ← θ
for m ← 1 to M do
  Experience Collection:
    Run current policy Ω_θ to collect a set of trajectories D
    Process states s_t through the self-attention module to obtain Z_t
  Advantage Estimation:
    Compute GAE Â_t and discounted returns R_t for all transitions in D
    Normalize advantages Â_t for training stability
  Network Optimization:
  for k ← 1 to K do
    Sample a mini-batch of size B from D
    Compute probability ratio r_t(θ) = Ω_θ(a_t|s_t) / Ω_θold(a_t|s_t)
    Dual-Clip Objective Calculation:
    if Â_t ≥ 0 then
      L(θ) ← min( r_t(θ)Â_t, clip(r_t(θ), 1 − ε, 1 + ε)Â_t )
    else
      L(θ) ← max( min(r_t(θ)Â_t, clip(r_t(θ), 1 − ε, 1 + ε)Â_t), cÂ_t )
    end if
    Update θ via stochastic gradient ascent on L(θ)
    Update φ via gradient descent on MSE loss L(φ) = (V_φ(s_t) − R_t)^2
  end for
  Synchronize: Update old policy parameters θ_old ← θ
end for
return Ω_θ
```

an excessively large c may overly restrict the policy's ability to correct poor behaviors. In practice, we choose $c = 3$ based on the guidelines in prior work, which balances stability and learning agility under large-scale off-policy sampling. This mechanism ensures that even upon encountering a catastrophic action sampling, the agent's policy undergoes a controlled, corrective update rather than an erratic divergence. The complete operational loop of the TDC-PPO agent is summarized in Algorithm 1.

V. Simulation Results

This section evaluates the performance of the proposed TDC-PPO algorithm in the considered NOMA-MISO VLC system. Through extensive simulations, we demonstrate the agent's ability to autonomously maximize the covert transmission rate while navigating complex system constraints and channel uncertainties.

A. Simulation Setup and Parameter Configuration

We consider a representative indoor VLC environment with dimensions of $6\,\mathrm{m} \times 6\,\mathrm{m} \times 4.5\,\mathrm{m}$. The system comprises $L = 6$ LED luminaires symmetrically deployed on the ceiling to provide uniform coverage. The precise coordinates of the luminaires are configured at $(1.5, 2, 3)\,\mathrm{m}$, $(3, 2, 3)\,\mathrm{m}$, $(4.5, 2, 3)\,\mathrm{m}$, $(1.5, 4, 3)\,\mathrm{m}$, $(3, 4, 3)\,\mathrm{m}$, and $(4.5, 4, 3)\,\mathrm{m}$.

TABLE I
Simulation Parameters

Description	Symbol	Value
Number of LEDs	L	6
Optical filter gain	g_{of}	1
Physical area of a PD	A_r	$1.0\,\mathrm{cm}^2$
Lambertian emission order	m	1.5
Minimum QoS requirement	$R_{\mathrm{c}}^{\mathrm{th}}$	$0.8\,\mathrm{bps/Hz}$
Minimum permissible current	$I_{\min}$	$0\,\mathrm{A}$
Maximum permissible current	$I_{\max}$	$4.8\,\mathrm{A}$
LED DC-bias current	I_{DC}	$2.4\,\mathrm{A}$
Maximum optical power per LED	$P_{\max}$	$35\,\mathrm{W}$
Noise power spectral density	σ^2	$-75\,\mathrm{dBm}$
Channel uncertainty factor	κ	0.2
Required covertness level	ϵ	0.05

To ensure the generality of our findings, the positions of the two legitimate users (Bob and Carol) and the monitoring user (Willie) are randomly distributed across the floor level. The Alice-agent must therefore learn a generalized policy capable of adapting to various channel realizations. The detailed simulation parameters, reflecting typical VLC hardware characteristics and covert communication requirements, are summarized in Table I. Unless otherwise specified, these parameters serve as the baseline configuration for all subsequent performance evaluations.

B. Impact of Hyper-Parameters on Convergence

In deep reinforcement learning, the learning rate (LR) is a pivotal hyper-parameter that governs the step size of parameter updates, directly influencing the trade-off between training pace and stability. An excessively large LR can induce high variance in updates, leading to training instability and preventing the convergence of the average reward. Conversely, an overly small LR may result in impractically slow convergence, potentially causing the agent to under-perform within a limited training budget.

Fig. 3 illustrates the convergence behavior of the TDC-PPO agent under different LR configurations. A learning rate of 10^{-4} results in a markedly sluggish growth in the average reward, confirming the drawback of insufficient step sizes. While a higher rate of 3×10^{-4} achieves the most rapid initial ascent, it subsequently induces significant oscillations in the reward curve, indicative of training instability caused by overly aggressive updates. In contrast, an LR of 2.5×10^{-4} achieves a superior balance, characterized by a smooth convergence trajectory and the highest final average reward. Consequently, 2.5×10^{-4} is selected as the optimal learning rate for all subsequent experiments.

C. Comparative Performance Analysis

To demonstrate the superiority of the proposed TDC-PPO algorithm in solving the constrained covert VLC optimization

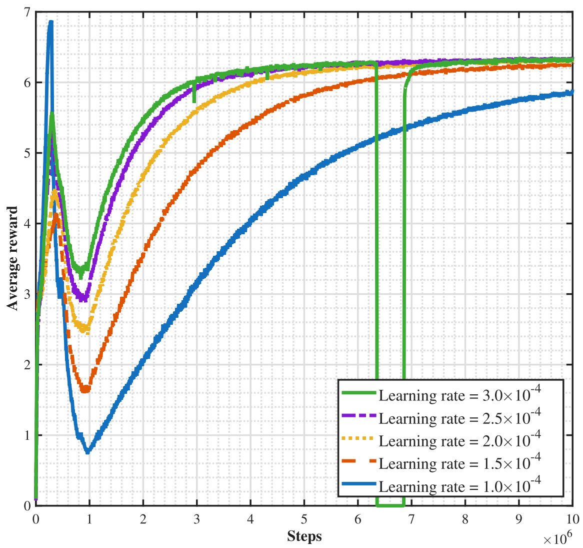

Fig. 3. Convergence performance under various learning rates.

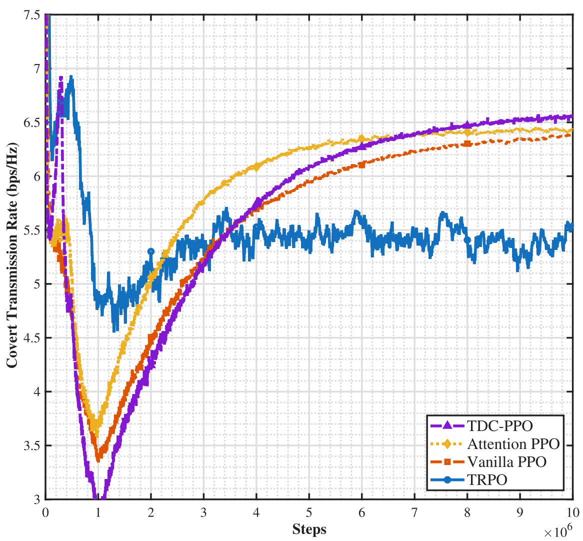

Fig. 4. Average reward comparison across different DRL algorithms.

problem, we compare its performance against three widely adopted baseline methods:

- **Vanilla PPO**: The standard PPO framework with on-policy sampling as proposed in [41].
- **Attention-PPO**: A PPO variant incorporating an attention mechanism for state encoding [42].
- **TRPO**: The trust region policy optimization method [43], known for its monotonic improvement guarantees.

As illustrated in Fig. 4, the proposed TDC-PPO agent significantly outperforms all baseline methods in terms of both convergence rate and final average reward. This superior performance is attributed to three synergistic factors: 1) the transformer-based architecture provides enhanced situational awareness to decouple NOMA interference from detection uncertainty; 2) the GAE mechanism effectively stabilizes training by balancing variance and bias in policy-gradient estimations; and 3) the dual-clip mechanism establishes a "safety floor" that prevents policy divergence caused by the harsh negative rewards associated with covertness violations.

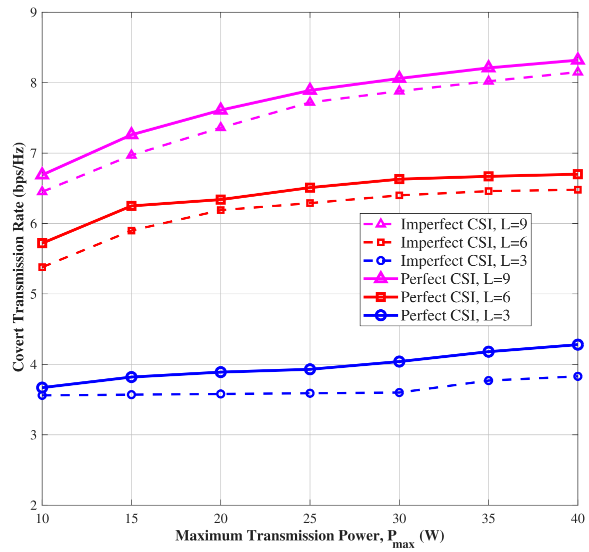

Fig. 5. Covert transmission rate versus $P_{\max}$ for varying LED counts.

Consequently, the Alice-agent achieves a more robust and efficient exploration of the optimal action space compared to conventional DRL methods.

D. Impact of Transmit Power and Spatial Diversity

Fig. 5 illustrates the covert transmission rate versus $P_{\max}$ under varying LED counts and CSI conditions. The covert rate increases with $P_{\max}$ but eventually reaches a saturation point beyond 25 W, where the Alice-agent intelligently limits power to avoid violating LED linearity or increasing Willie's detection probability. Furthermore, an increase in the number of LEDs enhances spatial degrees of freedom, allowing the agent to more effectively mask signals with AN. Notably, the narrowing performance gap between perfect and imperfect CSI as LED counts increase demonstrates the agent's ability to exploit spatial diversity to compensate for channel uncertainty, confirming its robustness in practical scenarios.

E. Adaptive Trade-Off Between Covertness and Rate

The relationship between the covert transmission rate and the covertness constraint ω is depicted in Fig. 6. As ω increases (relaxing the security requirement), the transmission rate grows because the agent autonomously reallocates power from masking mechanisms (AN and public signals) to the covert message stream. This growth saturates when other physical bottlenecks, such as the total power budget and LED dynamic range, become dominant. Similar to the power analysis, a larger LED array provides finer spatial control over optical energy, enabling the agent to maintain superior covert throughput across all security levels. This adaptive behavior validates that the TDC-PPO agent has successfully internalized the complex trade-offs between security thresholds and system capacity.

F. Impact of Deployment Height

Fig. 7 depicts the impact of deployment height on the agent's performance. As the LED height increases, the vertical

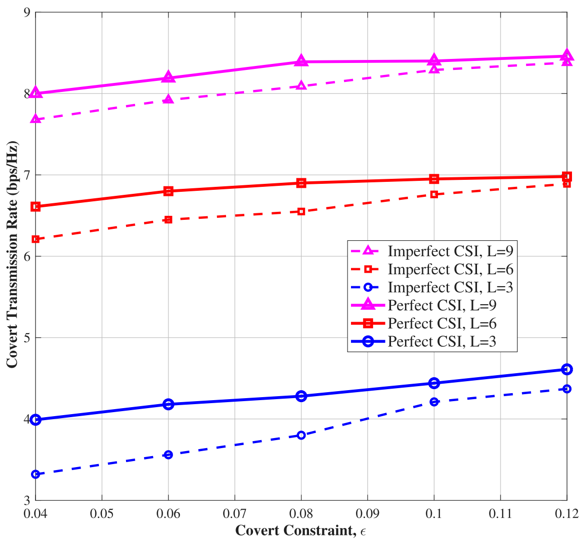

Fig. 6. Covert transmission rate versus covertness constraint ω.

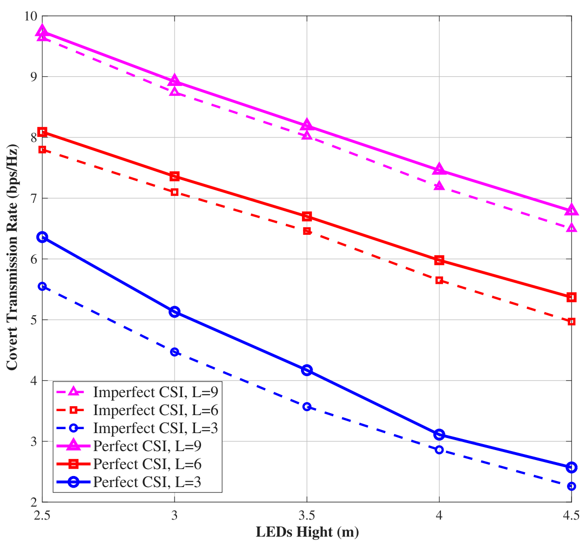

Fig. 7. Covert transmission rate versus LED deployment height.

distance to the users grows, leading to more pronounced path loss and a corresponding decline in the covert transmission rate across all configurations. In response to these degraded channel conditions, the Alice-agent demonstrates intelligent situational awareness by dynamically prioritizing signal energy toward Bob's coordinates. Simultaneously, it shapes the AN beamforming vectors to more precisely interfere with Willie's reception, thereby mitigating the performance loss caused by the weakened line-of-sight components. For the case with three LEDs, the performance gap between perfect and imperfect CSI narrows at greater heights, as the overall channel degradation due to path loss gradually dominates the effect of CSI uncertainty. Under heightened uncertainty, the agent's learned policy becomes increasingly risk-aware, adopting a more conservative power allocation strategy to ensure covertness even when the adversarial channel estimate is unreliable. This highlights the agent's robust decision-making capability in maintaining secure links despite unfavorable geometric deployment conditions.

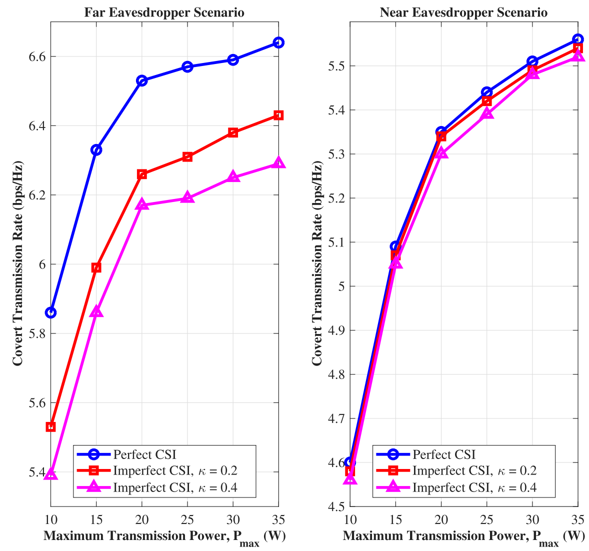

Fig. 8. Performance comparison under near-Eve and far-Eve scenarios.

G. Influence of Eavesdropper Proximity

Fig. 8 evaluates the covert transmission rate versus $P_{\max}$ for far-Eve and near-Eve scenarios under varying CSI conditions. When Willie is in close proximity to Bob, the high spatial correlation between their channel vectors $\mathbf{h}_{\rm b}$ and $\mathbf{h}_{\rm w}$ creates a severe "security bottleneck", as any signal intended for Bob is inevitably received with high gain by Willie. To counter this, the agent is forced to execute a highly precise beamforming policy that significantly restricts Bob's power share to maintain covertness, resulting in a lower rate compared to the far-Eve case. All curves exhibit a rapid initial rise followed by saturation at higher power levels. In the low $P_{\max}$ regime, the total power budget is the primary constraint, starving both AN and covert streams. As power increases, LED linearity and the absolute covertness threshold become the dominant limiting factors. The performance gap between far and near scenarios widens at high $P_{\max}$ because the agent can more effectively exploit the spatial separation in the far-Eve case to maximize resource utilization efficiency.

H. Resource Allocation Under Competing QoS Demands

Fig. 9 examines the agent's resource allocation behavior when faced with competing QoS demands from the public user, Carol. As Carol's QoS threshold $R_{\rm c}^{\rm th}$ rises, the Alice-agent is compelled to reallocate a larger fraction of the total power budget to the public signal, effectively siphoning resources away from the covert stream and the masking AN. This illustrates the agent's inherent trade-off reasoning; it must prioritize the "visible" service guarantee to maintain system integrity before optimizing for "hidden" covert performance. Under a tight power budget, the agent prioritizes Carol's QoS almost exclusively, leaving minimal residual power for covert transmission. However, as $P_{\max}$ increases, the agent learns to distribute the surplus power between AN sequences and Bob's beamforming vectors to maximize the covert rate without violating the strictly enforced security and hardware constraints. This adaptive behavior demonstrates the agent's capability for

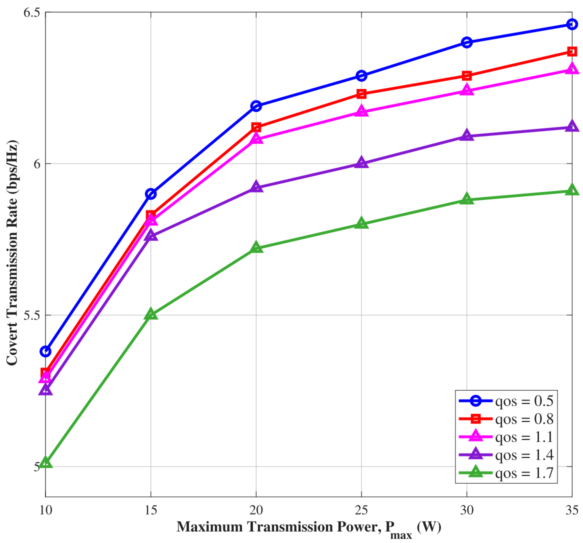

Fig. 9. Covert transmission rate versus $P_{\max}$ under varying QoS thresholds.

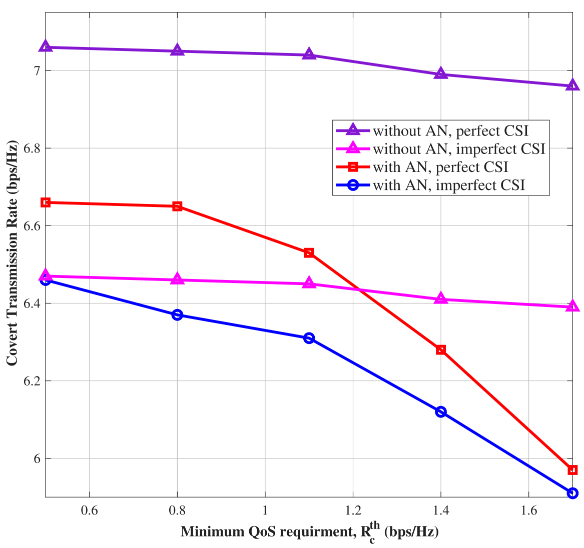

Fig. 10. Covert transmission rate vs. QoS threshold with and without AN.

sophisticated multi-objective optimization, dynamically adjusting its power-sharing policy to balance the conflicting goals of service reliability and transmission secrecy in a resource-constrained environment.

I. Impact of an Availability and QoS Constraints

Fig. 10 quantifies the joint influence of AN availability and QoS requirements on the covert transmission rate, revealing fundamental differences in the agent's strategic response. Without AN, the agent only has two signal streams to manipulate. When Carol's QoS requirement becomes stricter, the agent boosts the public signal power; although this increases interference at Willie, it cannot fully compensate for the power diverted from Bob, leading to a gradual rate decline. In contrast, when AN is available, the agent manages a three-way power distribution. Because AN is highly power-intensive yet essential for masking covert transmissions, even a modest increment in Carol's QoS entitlement markedly erodes

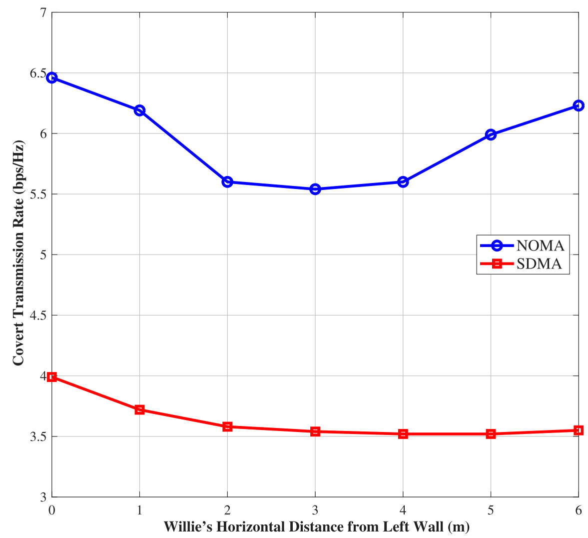

Fig. 11. Covert transmission rate versus Willie's horizontal position.

the resources available for effective secrecy. This results in a more pronounced monotonic drop in the covert rate as Carol's demands grow. These results confirm that while AN significantly enhances the absolute covert rate, it also increases the system's sensitivity to QoS fluctuations, a complex trade-off that the TDC-PPO agent successfully learns to navigate through iterative interaction with the environment.

J. Spatial Robustness and Multiple Access Comparison

Fig. 11 evaluates the covert transmission rate as a function of Willie's horizontal position, comparing the proposed NOMA-based agent against a traditional space-division multiple access (SDMA) scheme. The NOMA-based agent achieves peak performance when Willie is positioned near the room boundaries, where the large-scale fading and spatial separation are most favorable. However, the rate drops sharply as Willie approaches Bob's fixed central position, as the agent must redirect substantial power into masking interference to satisfy the intensified covertness constraints. This adaptive behavior is enabled by the NOMA action space, which allows the agent to dynamically allocate power among the public user, the covert user, and the artificial noise, while adhering to the successive interference cancellation ordering constraints. In contrast, the SDMA scheme maintains a nearly flat but significantly lower covert rate regardless of Willie's location. Under SDMA, users are isolated in orthogonal resource dimensions, and the agent lacks the flexibility to repurpose public user power as a dynamic cover for the covert link; the action space consists of fixed per-user power allocations without power-domain multiplexing, which limits the agent's ability to adapt to the eavesdropper's proximity. While SDMA offers strong spatial robustness, its poor spectral efficiency and inability to exploit power-domain multiplexing severely limit its effectiveness in high-security VLC scenarios. This comparison highlights that the TDC-PPO agent's policy is highly context-aware and spatially adaptive. By leveraging NOMA, the agent can dynamically reshape its energy distribution based on the adversary's location, offering superior performance over the static and inefficient resource allocation characteristic of traditional SDMA.

VI. Conclusion

This paper proposed an agentic AI framework for autonomous covert communication in MISO-NOMA VLC systems under imperfect CSI. By modeling the transmitter as an intelligent Alice-agent that optimizes power allocation and beamforming through a novel TDC-PPO algorithm, we achieved robust covertness and efficient transmission rates without requiring prior knowledge of the eavesdropper's channel. Simulation results confirm that this agentic approach outperforms standard PPO and TRPO baselines in terms of convergence stability and situational adaptability. The agent successfully navigates the complex trade-offs between public QoS requirements, LED hardware limits, and covertness constraints. By embedding a transformer-based architecture and a dual-clip mechanism, the framework demonstrates a strong capability to maintain secure communication links in various indoor deployment scenarios. This work validates the effectiveness of learning-driven agents in enhancing physical-layer security within self-optimizing VLC networks. Building on these results, future research can extend the proposed framework to multi-cell VLC systems to address inter-cell interference and cooperative resource allocation, as well as integrate with radio frequency networks to leverage the complementary strengths of both communication modalities for improved coverage and reliability.

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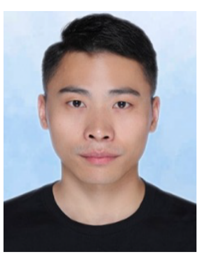

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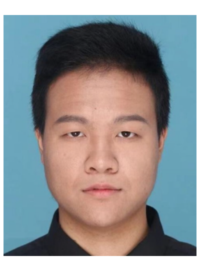

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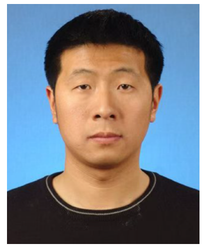

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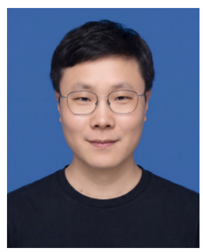

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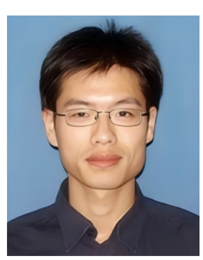

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